Big data in combinatorics: is it FAIR?

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Overview

- FAIR data principles
- The OpenDreamKit project
- Some datasets
- Work in progress

The FAIR data principles

In 2014, a Lorentz Center (Leiden) workshop began formulating the FAIR Data Principles for research data. These principles were elaborated over the following two years.

FAIR stand for Findable, Accessible, Interoperable, Reusable

(Aalbersberg et al. 2014, Wilkinson et al. 2016)

To be Findable

F1 (meta)data are assigned a globally unique and persistent identifier

F2 data are described with rich metadata (defined by R1)

F3 metadata clearly and explicitly include the identifier of the data it describes

F4 (meta)data are registered or indexed in a searchable resource

(Wilkinson et al. 2016)

To be Accessible

A1 (meta)data are retrievable by their identifier using a standardized communications protocol

A1.1 the protocol is open, free, and universally implementable

A1.2 the protocol allows for an authentication and authorization procedure, where necessary

A2 metadata are accessible, even when the data are no longer available

(Wilkinson et al. 2016)

To be Interoperable

I1 (meta)data use a formal, accessible, shared, and broadly applicable language for knowledge representation.

I2 (meta)data use vocabularies that follow FAIR principles

I3 (meta)data include qualified references to other (meta)data

(Wilkinson et al. 2016)

To be Reusable

- **R1** meta(data) are richly described with a plurality of accurate and relevant attributes
- **R1.1** (meta)data are released with a clear and accessible data usage license
- **R1.2** (meta)data are associated with detailed provenance
- **R1.3** (meta)data meet domain-relevant community standards

(Wilkinson et al. 2016)

The FAIR data principles in Australasia

- Australian FAIR access working group

  F.A.I.R. Access Policy Statement
  "By 2020, Australian publicly funded researchers and research organisations will have in place policies, standards and practices to:

  1. Make publicly funded research outputs findable, accessible, interoperable and reusable . . . "

- Strategic Science Investment Fund (New Zealand)

  New Zealand Data Science Research Programmes Call for Proposals
  "Describe the excellence of your team by telling us: . . . How you will access, analyse and share data, via appropriate systems and tools, including implementation of the FAIR data principles for scientific data management and stewardship. . . . "

(https://www.fair-access.net.au/; https://www.mbie.govt.nz/)

The OpenDreamKit project

The EU funded OpenDreamKit project (2015-2019) included a census of mathematical data sets, and the development of prototypical semantic infrastructure for the FAIR sharing of such data sets.

This used the idea of "Deep FAIR": FAIR sharing at the level of mathematical objects.

(OpenDreamKit Project 2015-2019)

Semantics of concrete data: an example

Graphs, up to isomorphism can be stored using a canonical labelling in Graph6 format.

Different Graph6 strings represent non-isomorphic graphs, *as long as the same canonical labelling algorithm is used*.

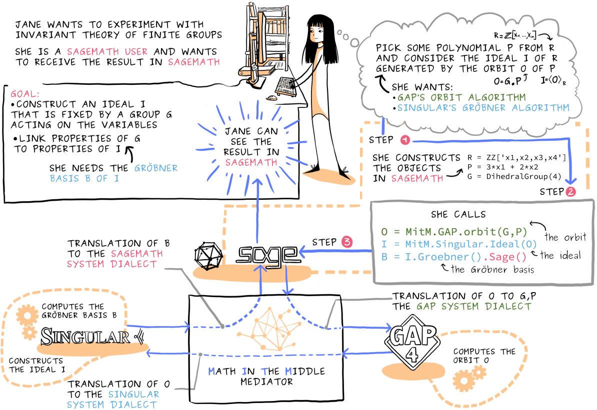
In SageMath 8.6, the `"bliss"` and `"sage"` algorithms yield *different* canonical labels for the same graph.

Thus some sort of semantic metadata needs to be stored in long term graph databases, to avoid having to canonically label again.

(SageMath 2019)

Semantic infrastructure in OpenDreamKit

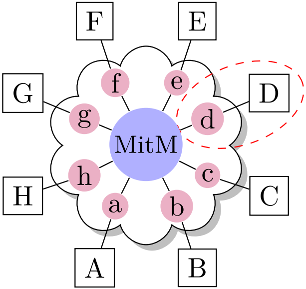
Math in the Middle, MMT

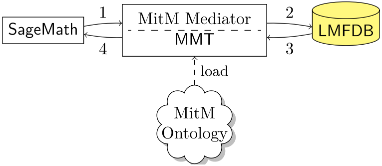


(OpenDreamKit Project 2015-2019; Kohlhase 2018)

Semantic infrastructure in OpenDreamKit

Math in the Middle, MMT, virtual theories





(OpenDreamKit Project 2015-2019)

The FAIRmat project proposal

The FAIRmat project was intended to be a follow-on from the OpenDreamKit project, concentrating on the furter development of FAIR infrastructure for mathematical software and datasets, within the European Open Research Cloud.

The project was intended to begin in 2020, but was not funded.

(FAIRmat Project Proposal, Kohlhase 2019)

Datasets of bent functions

- Langevin's lists of partial spread bent functions.
  - Flat text files.
  - Encoded by algebraic normal form.
- Langevin's "Classification of Boolean Quartics Forms in eight Variables."
  - Flat text files.
  - Encoded by algebraic normal form.
  - Does not explicitly enumerate bent functions or their equivalence classes.

(Langevin 2010; Langevin 2008)

Databases of strongly regular graphs

- Spence's lists of strongly regular graphs on at most 64 vertices.
  - Exhaustive lists of non-isomorphic graphs for some tuples of feasible parameters.
  - Flat text files.

- Brouwer's database of parameters of strongly regular graphs.

- Cohen and Pasechnik's Sage implementation of Brouwer's database, with constructions.
  - Example for each tuple of feasible parameters, if known.
  - Sage interface.

(Spence 1995-2006; Brouwer 2008-2017; Cohen and Pasechnik 2016-2017)

Other recent mathematical datasets

- House of Graphs: a databse of interesting graphs
  - "...The key principle is to have a searchable database and offerânext to complete lists of some graph classes – also a list of special graphs that have already turned out to be interesting and relevant in the study of graph theoretic problems or as counterexamples to conjectures."
  - Web-based interface.

- LMFDB - The L-functions and modular forms database
  - "The LMFDB is an extensive database of mathematical objects arising in Number Theory."
  - Based on MongoDB and Python.
  - Web-based and Sage interfaces.

(Brinkmann et al. 2013; The LMFDB Collaboration 2007-2017)

Bent function Cayley graph databases

Three prototype relational databases, using SQLite:

- Bent functions in 6 dimensions.
  - Database of 11 strongly regular graphs from 4 ET classes.
  - Database size 780 KB.
- Bent functions in 8 dimensions, up to degree 3.
  - Database of 55 strongly regular graphs from 9 ET classes.
  - Database size 65 MB.
- Bent functions from the 8 S-boxes of CAST-128.
  - Database of more than *32 million* $(32\,914\,496)$ strongly regular graphs from 256 ET classes and their duals.
  - Database size *195 GB*.

Flat directories of SageMath sobj files:

About *2 billion* strongly regular graphs from the 14 758 ET classes of partial spread bent functions in 8 dimensions and their duals. *Over 6 TB* total size.

Schema for Cayley graph databases

Name	Type	Schema
Tables (4)		
bent_function		CREATE TABLE bent_function(bent_function
bent_function	BLOB	`bent_function` BLOB
name	TEXT	`name` TEXT UNIQUE
cayley_graph		CREATE TABLE cayley_graph(bent_function I
bent_function	BLOB	`bent_function` BLOB
cayley_graph_index	INTEGER	`cayley_graph_index` INTEGER
graph_id	INTEGER	`graph_id` INTEGER
graph		CREATE TABLE graph(graph_id INTEGER, ca
graph_id	INTEGER	`graph_id` INTEGER
canonical_label_hash	BLOB	`canonical_label_hash` BLOB UNIQUE
canonical_label	TEXT	`canonical_label` TEXT
matrices		CREATE TABLE matrices(bent_function BLOE
bent_function	BLOB	`bent_function` BLOB
b	INTEGER	`b` INTEGER
c	INTEGER	`c` INTEGER
bent_cayley_graph_index	INTEGER	`bent_cayley_graph_index` INTEGER
dual_cayley_graph_index	INTEGER	`dual_cayley_graph_index` INTEGER
weight_class	INTEGER	`weight_class` INTEGER

Database schema for SQLite version of the classification database.

Possible use case: Roberson's classification of strongly regular graphs

Roberson recently proved that all strongly regular graphs are cores or have complete cores, proposing four types, A, B, C, X according to clique number and chromatic number.

Question: for each dimension of bent functions, determine the number of strongly regular Cayley graphs of each type.

(Roberson 2019)

Bent function Cayley graph Sage code

CoCalc: Public worksheets, Sage and Python source code

http://tinyurl.com/Boolean-Cayley-graphs

GitHub: Sage and Python source code

https://github.com/penguian/Boolean-Cayley-graphs

PyPI: Installable Python package

https://pypi.org/project/boolean-cayley-graphs/

SourceForge: Documentation of Python modules

https://boolean-cayley-graphs.sourceforge.io/

Possible next steps

- Rewrite and resubmit "Classifying bent functions by their Cayley graphs."
- Develop a RESTful API for the bent function Cayley graph databases.
- Develop semantic interfaces using Math-in-the-Middle, and other OpenDreamKit services.
- Arrange for hosting. EUDAT? Australian Research Data Commons?
- Scale up to the partial spread bent functions?